

REVIEW ARTICLE

ChatGPT in Education: A Systematic Review of Current Landscape, Limitations and Future Directions Through General System Theory Lens

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ABSTRACT

The rapid proliferation of ChatGPT has fuelled expectations of transformative change in education; yet most systematic reviews remain confined to single disciplines or educational levels. Adopting general system theory (GST) and the knowledge-skills-attitudes (KSA) framework, this review offers a more comprehensive perspective on ChatGPT's educational impact. Following PRISMA guidelines, we searched major academic databases for studies published between November 2022 and March 2024, screened them against predefined criteria, and ultimately analysed 92 eligible papers through qualitative content analysis. Findings were mapped onto the five GST elements—subject, information, medium, environment and technology. Research to date is markedly concentrated in higher education (69.7%), with far less work in primary (3.4%) and secondary (13.5%) settings. Nearly half of the studies target STEM subjects (45.6%). The most frequently reported applications are timely feedback (55 occurrences), content generation (20) and personalised learning (15). Positive effects are most evident for knowledge construction (63 occurrences) and skills development (59), whereas attitudinal outcomes receive comparatively little attention (20). Overall, the evidence points to promising—but uneven—educational benefits. Prevailing limitations include small- or medium-scale research designs and the under-representation of primary and secondary contexts. Future work should adopt larger samples, give greater weight to attitudinal dimensions, and strive for a more balanced coverage across all educational levels to realise ChatGPT's full potential in teaching and learning.

1 | Introduction

The rapid advancement of natural language processing (NLP) has ushered in a new era of educational technology, with ChatGPT emerging as a transformative force in teaching and learning environments (Ray 2023). This large language model,

developed by OpenAI, has demonstrated remarkable capabilities in generating human-like responses, sparking widespread interest in its potential to revolutionise educational practices across various disciplines and age groups (Chiu 2024; Crompton and Burke 2023; Wu et al. 2024). This phenomenon is not confined to a single region; educational systems globally

Abbreviations: AI, artificial intelligence; API, application programming interface; GST, general system theory; KSA, knowledge, skills and attitudes; LLM, large language model; NLP, natural language processing; PRISMA, preferred reporting items for systematic reviews and meta-analyses; STEM, science, technology, engineering and mathematics.

are grappling with how to best integrate this technology, leading to a diverse and sometimes fragmented international landscape of research and practice that necessitates a comprehensive synthesis.

Recent research has highlighted the positive impact of integrating ChatGPT with activity-based learning approaches on students' performance in education (Al Shloul et al. 2024; Li 2023; Park and Kim 2025), further underscoring its potential to enhance educational outcomes. However, the integration of ChatGPT into educational settings presents both opportunities and challenges that require careful examination and systematic analysis.

While existing research has explored ChatGPT's applications in specific educational contexts, there is a notable lack of comprehensive studies that analyse its impact across diverse settings and learning outcomes. Previous reviews have often focused on particular disciplines or educational levels, such as higher education (Lee and Wu 2025; Zitar 2023) or K-12 education (Zhang and Tur 2023), without providing a holistic understanding of ChatGPT's potential to transform teaching and learning practices across the entire educational spectrum. Moreover, the rapid evolution of ChatGPT and its applications in education has outpaced the existing literature, leaving educators and policymakers without a comprehensive, up-to-date synthesis of its effects on various aspects of student development.

The lack of a systematic examination of ChatGPT's impact on knowledge construction, skill development and attitudinal changes across different age groups, disciplines and learning environments hinders our ability to make informed decisions about its integration into educational practices. This challenge is compounded by the need to address cybersecurity issues in education when implementing AI technologies like ChatGPT (Mazhar et al. 2023), ensuring the protection of student data and privacy. Additionally, the absence of a unifying theoretical framework in previous studies has resulted in fragmented insights that fail to capture the complex interactions between ChatGPT and various elements of the educational ecosystem. This study is motivated by the urgent need to address these limitations and provide a comprehensive, theoretically grounded analysis of ChatGPT's applications and impacts in education.

To address these gaps and provide a comprehensive, theoretically grounded analysis of ChatGPT's applications and impacts in education, this study employs general system theory (GST) as a theoretical framework. GST, introduced by Bertalanffy (1968), offers a holistic approach to understanding complex systems as interconnected elements that dynamically interact to achieve a common purpose. This framework has been successfully applied in various fields to analyse complex systems and their interactions.

In educational research, GST has proven valuable in examining the multifaceted nature of learning environments and technologies. For instance, Chen and Stroup (1993) used GST to develop a framework for science education, emphasising the interconnectedness of scientific concepts and processes. The inherent

characteristics of GST—particularly its focus on holistic analysis, dynamic interactions, emergent properties and system boundaries—make it exceptionally well suited for studying AI technologies in education (Memarian and Doleck 2025; Rousseau 2015). AI systems like ChatGPT function as complex, adaptive subsystems within larger educational ecosystems, with inputs (prompts, contextual data), transformation processes (computational analysis, pattern recognition) and outputs (generated responses, feedback) that closely mirror the systemic elements central to GST (Holmes and Tuomi 2022; Salih et al. 2025). In the context of emerging technologies, Xu and Ouyang (2022) employed GST to conduct a systematic review of AI technologies in STEM education, demonstrating the framework's efficacy in capturing the complex, bidirectional interactions between AI tools, learners and educational environments. Their analysis revealed how GST's emphasis on interconnectedness helps explain the emergent learning effects that arise from AI integration in educational settings—effects that cannot be understood by examining individual components in isolation. More recently, Jia et al. (2024) applied GST to analyse the effectiveness of educational technologies, highlighting how AI technologies represent open systems that continuously exchange information with their environments, adapting to user inputs while simultaneously influencing the learning context—a fundamental principle of GST. Similarly, Glickman and Sharot (2025) demonstrated how GST provides an invaluable conceptual framework for understanding the complex feedback loops between AI learning systems, human learners and instructional contexts, allowing researchers to identify both intended and unintended consequences of AI integration in education.

By applying GST to the analysis of ChatGPT in education, we aim to provide a comprehensive understanding of how this AI technology interacts with and influences various components of the educational ecosystem, including subjects, information, mediums, environments and technologies. This approach is informed by recent efforts to synthesise educational theories with emerging technologies for enhanced learning experiences (Mallek et al. 2024; Watson and Romic 2024), suggesting potential benefits in applying a similar synthetic approach to ChatGPT in education.

Furthermore, this study incorporates the Knowledge, Skills and Attitudes (KSA) framework to evaluate ChatGPT's impact on learning outcomes. This approach allows for a multidimensional assessment of ChatGPT's educational value, considering cognitive, psychomotor and affective domains of learning (Bloom et al. 1956; Krathwohl 2002). The integration of ChatGPT into educational practices may also necessitate new approaches to analysing enrolment criteria and student performance, as highlighted by recent research applying machine learning and data mining techniques in educational contexts (Abideen et al. 2023; Kanbul et al. 2024; Park and Kim 2025).

Guided by these frameworks, our study seeks to answer two fundamental research questions:

1. What is the distribution and types of ChatGPT applications in relation to the five key elements of the GST framework (subject, information, medium, environment and technology) in education?

2. How do the other four key elements of the GST framework (subject, information, medium and environment) influence the technology element in the context of ChatGPT applications in education?

By addressing these questions, we aim to provide a nuanced understanding of how ChatGPT interacts with and influences various components of the educational ecosystem. Examining the distribution and types of ChatGPT applications across the GST elements will help identify patterns and trends in its current use, highlighting areas of concentration and potential gaps in implementation. Furthermore, investigating the interplay between the technology element and the other GST elements will uncover the complex relationships and dependencies that shape the effectiveness and impact of ChatGPT in educational settings.

This systematic review contributes to the existing body of knowledge in several ways. First, it provides a novel application of GST to the analysis of AI in education, offering a more nuanced understanding of the complex interactions between ChatGPT and various elements of the educational system. Second, it presents a comprehensive review of ChatGPT applications across different educational levels, disciplines and contexts, addressing the limitations of previous studies that focused on specific areas. Third, by employing the KSA framework, this study offers a multidimensional evaluation of ChatGPT's impact on learning outcomes, providing a more holistic assessment of its educational value.

The findings of this study will offer valuable insights for educators, institutions and policymakers seeking to effectively leverage ChatGPT in diverse educational settings. By providing a comprehensive analysis of ChatGPT's integration into educational practices and its potential to influence teaching and learning outcomes, this review aims to inform evidence-based decision-making and guide future research in the rapidly evolving field of AI-enhanced education.

2 | Related Work

2.1 | ChatGPT in Education

ChatGPT has emerged as a powerful tool in education, offering unprecedented capabilities in NLP and generation (Zhang and Tur 2023). Its versatility allows it to adapt to various educational disciplines and age groups, making it particularly well-suited for applications ranging from language learning (Dakakni and Safa 2023; Vázquez-Cano et al. 2023) to STEM education (AlGhamdi 2024; Lee et al. 2024). Based on the comprehensive review of existing literature and the potential applications of ChatGPT in educational settings (Jeon et al. 2023; Wu et al. 2024), this study categorises ChatGPT's educational functions into four primary areas: automated assessment, personalised learning, timely feedback and content generation.

Automated assessment leverages ChatGPT's natural language understanding capabilities to evaluate student responses, potentially streamlining the grading process and providing consistent, objective feedback (Jukiewicz 2024; Steiss et al. 2024). This function addresses the growing need for efficient assessment

methods in increasingly large and diverse educational environments. Automated assessment through ChatGPT represents a significant advancement over traditional automated grading systems, which typically rely on pre-programmed rubrics and pattern matching (Martínez-Huertas et al. 2019). ChatGPT's capacity to understand semantic meaning, context and nuance enables it to evaluate complex student work such as essays, short answers and creative projects with greater sophistication (Bui and Barrot 2025). Recent studies have demonstrated ChatGPT's effectiveness in providing assessment comparable to human graders across multiple subjects, including mathematics (Gandolfi 2025), language arts (Steiss et al. 2024) and science (Covington and Vruwink 2024). For instance, Pinto et al. (2023) found that ChatGPT demonstrated substantial agreement with expert graders when evaluating open-ended computer science questions, suggesting its potential to reduce instructor workload while maintaining assessment quality. The educational value of automated assessment extends beyond efficiency, as it can increase assessment frequency, provide more immediate learning interventions, and enable instructors to focus on higher-value pedagogical activities such as curriculum design and student mentoring (Bower et al. 2024; Küchemann et al. 2023).

Personalised learning harnesses ChatGPT's ability to adapt its responses based on individual learner characteristics and needs (Monib et al. 2025). This function aligns with the growing emphasis on tailored educational experiences and recognises the diverse learning styles and paces of students. ChatGPT achieves personalisation through several mechanisms: contextual understanding of user queries, memory of conversation history and adaptability to different knowledge levels and learning needs (Jauhiainen and Guerra 2023). Unlike traditional adaptive learning systems that typically follow pre-determined pathways, ChatGPT can generate unique explanations, examples and learning activities tailored to individual students' questions, interests and prior knowledge (Lee et al. 2024; Wu et al. 2024). Recent research has demonstrated how this personalisation capability supports differentiated instruction for diverse learners, including those with special educational needs (Xu et al. 2024), second language learners (Moorhouse 2024) and students requiring remedial support (Chang et al. 2023). For example, Li (2023) found that ChatGPT-enhanced flipped learning significantly improved learners' project performance by providing personalised guidance that addressed individual misconceptions and knowledge gaps. The educational value of such personalisation extends beyond content mastery to include increased learner engagement, improved self-efficacy and enhanced motivation through experiences that are relevant and appropriately challenging for each student (Alamri et al. 2020; Fariani et al. 2023; Wu et al. 2023).

Timely feedback, a critical aspect of effective learning, is facilitated by ChatGPT's capacity to provide instant responses and guidance (Fung et al. 2023; Wu et al. 2024). This immediacy can help address learners' questions or misconceptions in real time, preventing the accumulation of knowledge gaps and maintaining engagement. The timely feedback function of ChatGPT represents a significant advancement in educational technology, as it overcomes limitations of traditional feedback mechanisms that often involve delays between student performance and corrective information (Huang et al. 2025; Wu

et al. 2024). Drawing on principles of formative assessment and scaffolding (Black and Wiliam 2009; Vygotsky and Cole 1978), ChatGPT provides not only verification feedback (correct/incorrect) but also elaborate feedback with explanations, examples and guidance for improvement (Lee et al. 2024). Recent empirical studies have demonstrated the effectiveness of ChatGPT's feedback across diverse educational contexts, including programming education (Kazemitabaar et al. 2024), academic writing (Nguyen et al. 2024) and problem-solving activities (Lee et al. 2024). For instance, Meyer et al. (2024) found that AI-generated feedback significantly increased secondary students' text revision behaviours, motivation and positive emotions compared to traditional feedback methods. The immediacy and quality of this feedback address a persistent challenge in education where instructor capacity to provide detailed, individualised feedback is often limited by time constraints and large class sizes (Lee, Lin, et al. 2023). Moreover, the continuous availability of ChatGPT feedback extends learning support beyond formal class hours, potentially transforming how and when students receive guidance in their educational journey (Wang et al. 2025; Yan 2025).

Content generation utilises ChatGPT's language generation capabilities to create educational materials, potentially assisting educators in developing engaging and diverse learning resources (Han et al. 2024; Silva et al. 2024). This function acknowledges the time-intensive nature of content creation and the potential for AI to support this process. ChatGPT's content generation capabilities extend beyond simple text production to include creating tailored learning materials such as lecture notes, study guides, discussion prompts, case studies and practice problems (Kiyak and Kononowicz 2024). Drawing on its vast training corpus of texts across numerous domains, ChatGPT can generate materials at varying complexity levels, in different formats, and from multiple perspectives (Einarsson et al. 2024). Recent studies have demonstrated how this capability supports instructional design and curriculum development across disciplines, including language education (Liu and Xiao 2025), medical education (Cecchini et al. 2024) and STEM subjects (Bitzenbauer 2023). For example, Husain (2024) found that programming instructors utilising ChatGPT for content generation reported significant time savings and increased diversity in learning resources. The educational value of AI-assisted content generation extends beyond efficiency to include opportunities for creating more inclusive, culturally responsive and multimodal learning materials (ElSayary 2023). Additionally, the iterative nature of working with ChatGPT allows educators to rapidly prototype, refine and personalise learning resources to better meet the needs of specific student populations (Jauhiainen and Guerra 2023; Park and Choo 2024).

These four categories were identified based on their prevalence in the literature and their potential to address key challenges and opportunities in education. They represent distinct yet interconnected aspects of ChatGPT's educational applications, covering both student-facing (e.g., personalised learning, timely feedback) and teacher-facing (e.g., automated assessment, content generation) functionalities.

However, it is crucial to recognise that the effectiveness of these functions may vary significantly across age groups and

subject areas due to differences in prior knowledge, cognitive abilities and learning styles (Hu et al. 2023; Kasneci et al. 2023). This variability underscores the need for a comprehensive understanding of ChatGPT's impact on learners across diverse contexts to guide its effective and appropriate integration into educational practices.

By focusing on these four key functions, this study aims to provide a structured framework for analysing ChatGPT's applications in education, facilitating a more nuanced understanding of its potential benefits and limitations across different educational settings and learner populations.

2.2 | Knowledge, Skills and Attitudes (KSA)

The KSA framework, closely aligned with Bloom's taxonomy (Bloom et al. 1956), offers a comprehensive approach for categorising and assessing learning outcomes. The framework considers three main domains: cognitive (knowledge), psychomotor (skills) and affective (attitudes) (Krathwohl 2002).

Knowledge outcomes involve the acquisition and understanding of factual information, concepts, principles and theories (Anderson and Krathwohl 2001). In alignment with Bloom's Taxonomy, ChatGPT supports multiple levels of knowledge construction in STEM education. At the remembering level, ChatGPT assists learners in recalling fundamental scientific facts, mathematical formulas and technology concepts, providing on-demand clarifications that reinforce foundational knowledge (Wu et al. 2024). For the understanding level, ChatGPT generates explanations of complex STEM phenomena using multiple representations, analogies and examples tailored to individual learning needs (Chan et al. 2025; Wu et al. 2025). Research demonstrates that ChatGPT's ability to present STEM concepts in varied contexts enhances conceptual understanding and knowledge transfer across domains (Chen et al. 2025; Li et al. 2024). Particularly in technical subjects, ChatGPT facilitates the mastery of domain-specific terminology, theoretical frameworks and subject-specific mental models that form the cognitive foundation for advanced learning (Bhattacharya et al. 2024). When compared to traditional learning resources, ChatGPT's interactive nature enables a more dynamic and responsive knowledge construction process, allowing learners to actively refine their mental models through dialogue rather than passive consumption of information (Zhang 2023). In the context of ChatGPT applications in education, knowledge construction is supported through the provision of accurate information, explanations of complex concepts and engagement in meaningful discussions (Wu et al. 2024).

Skills outcomes refer to the ability to apply knowledge to perform specific tasks or solve problems (Simpson 1966). While the psychomotor domain traditionally focuses on physical abilities (Knecht 1974), in the context of ChatGPT, skills can include cognitive and linguistic abilities such as critical thinking, problem-solving, communication and collaboration (Lee et al. 2024). Within STEM education contexts, ChatGPT enhances skill development across various technical and applied competencies that align with the psychomotor domain adapted for digital learning environments. Research demonstrates that

ChatGPT scaffolds the development of programming capabilities by providing step-by-step guidance, debugging assistance and explanations of code functionality (Kazemitabaar et al. 2024). In data analysis applications, ChatGPT supports skills in data visualisation, statistical interpretation and experimental design through interactive problem-solving sessions (Alfirević et al. 2024; Huang et al. 2024). For communication skills essential in STEM disciplines, studies show ChatGPT improves technical writing proficiency, scientific reporting and presentation of complex information through guided practice and immediate feedback (Woolever and Wettstein 2024). In collaborative learning contexts, ChatGPT facilitates the development of teamwork skills through simulated discussions and project management support (Jony and Hamim 2024). Particularly notable in STEM research is ChatGPT's ability to bridge theoretical knowledge and practical application, enabling learners to translate abstract concepts into concrete solutions—a fundamental aspect of the psychomotor domain in technological education (Allen et al. 2024; Valeri et al. 2025). These skill-building capabilities directly correspond to measurable improvements in applied abilities that are consistently documented in empirical studies of ChatGPT integration in STEM education. ChatGPT supports skill development by providing opportunities to practise and apply knowledge through interactive exercises, simulations and real-world scenarios (Al Shloul et al. 2024; Rasul et al. 2023).

Attitudinal outcomes encompass the affective domain of learning, which includes feelings, values, appreciations, enthusiasms, motivations and attitudes (Krathwohl 2002). In alignment with Bloom's Taxonomy, ChatGPT influences the affective domain across various levels in STEM education. At the receiving level, ChatGPT creates low-stakes, non-judgmental learning environments that reduce anxiety and increase willingness to engage with challenging STEM concepts (Ng et al. 2024). For responding and valuing levels, research shows that ChatGPT's immediate, personalised responses foster increased curiosity, sustained motivation and greater self-efficacy in STEM subjects, particularly among learners who traditionally struggle with technical content (Gao et al. 2024; Mahande et al. 2025). At higher affective levels like organisation and characterisation, ChatGPT contributes to the development of scientific mindsets, intellectual perseverance and STEM identity formation by validating students' inquiries and encouraging iterative problem-solving approaches (Ilić et al. 2024; Valeri et al. 2025). The anthropomorphic qualities of ChatGPT create social presence and emotional engagement that traditional learning resources cannot achieve, facilitating affective connections to STEM subjects that have historically been perceived as impersonal or intimidating (Ackermann et al. 2025). This domain is essential for shaping learners' dispositions, beliefs and behaviours towards learning and their fields of study (Martin and Briggs 1986). ChatGPT can influence learners' attitudes by providing personalised feedback, fostering engagement and motivation and creating a supportive learning environment that encourages exploration and self-reflection (Cai et al. 2023; Lee, Lin, et al. 2023).

The KSA framework provides a structured lens to evaluate the impact of ChatGPT on learning outcomes, considering cognitive, psychomotor and affective dimensions. By examining how

ChatGPT influences knowledge construction, skill development and attitudinal changes, researchers can gain a comprehensive understanding of its educational value and potential to transform teaching and learning practices.

2.3 | Limitations of Existing Research

Existing systematic reviews on ChatGPT and AI applications in education offer valuable insights but have significant limitations. Zirar (2023) focused mainly on higher education, covering only 25 articles up to August 2023, resulting in a narrow research scope. Zhang and Tur (2023) reviewed ChatGPT applications in K-12 education based on 13 articles published up to June 2023, but while they identified several strengths of ChatGPT including personalised learning and positive influence on student achievement, their SWOT analysis approach did not systematically evaluate specific learning outcomes. Their review lacked a structured assessment framework examining how ChatGPT affects various dimensions of student development, such as knowledge acquisition (factual understanding, concept mastery), skills enhancement (critical thinking, problem-solving, language skills) and attitudinal changes (motivation, engagement) across different subject domains and age groups within K-12 education.

Crompton and Burke (2024) examined ChatGPT's impact on teaching and learning using 44 articles from November 2022 to August 2023 but lacked a systematic analysis across disciplines and age groups. Another study by Crompton and Burke (2023) reviewed AI applications in higher education, including 138 articles up to the end of 2022, which may not capture the latest developments post-ChatGPT release. Jeon et al. (2023) focused on language learning and speech recognition chatbots, with limited attention to ChatGPT, restricting their ability to assess its broader educational impact.

These reviews' limited number of articles and narrow focus on specific educational levels or time periods highlight the need for a comprehensive and up-to-date systematic review of ChatGPT applications in education. The absence of a thorough examination of ChatGPT's impact on various aspects of student development underscores the necessity of a multidimensional evaluation approach.

This study addresses these limitations by conducting a large-scale systematic review of ChatGPT applications across various age groups, disciplines and educational contexts. Using the GST framework, this review provides a holistic perspective on ChatGPT's interaction with different components of the educational ecosystem. Incorporating the KSA framework, it evaluates ChatGPT's impact on learning outcomes in cognitive, psychomotor and affective domains.

Employing rigorous methodological approaches, including clear inclusion and exclusion criteria, a comprehensive search strategy and a structured data extraction and synthesis process based on GST and KSA frameworks, this study distinguishes itself from previous reviews. By providing a transparent and systematic analysis of the available evidence, this review aims to generate reliable insights and recommendations for effectively integrating ChatGPT into educational practices.

3 | Methodology

To map the state-of-the-art applications of ChatGPT in education, a systematic review was conducted from November 2022 to March 2024, adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) principles (Moher et al. 2009). The selection process, grounded in PRISMA guidelines, is depicted in Figure 1.

3.1 | Database Search

A comprehensive search was conducted in major databases, including Web of Science, Science Direct, Scopus, IEEE, ACM, Taylor and Francis and Wiley (Guan et al. 2020; Xu and Ouyang 2022). The search strategy employed filters to isolate empirical research and peer-reviewed articles within the domain of education and educational research, spanning from November 2022—the month when ChatGPT was launched—until March 2024.

3.2 | Search Strategy

The search strategy was guided by the research questions and utilised three categories of keywords. The first category included terms related to ChatGPT and its applications in large language models, such as ‘generative artificial intelligence’, ‘GAI’, ‘ChatGPT’, ‘GPT’, ‘large language model’, and ‘LLM’. The second category focused on the educational domain, incorporating keywords like ‘education’, ‘learning’, ‘teaching’, ‘class’, and ‘course’. This approach ensured a thorough exploration of both technological and educational dimensions. Table 1 details the Boolean queries used in the keyword search.

3.3 | Inclusion and Exclusion Criteria

The inclusion criteria for the systematic review were: (a) focus on the application of ChatGPT in educational settings, (b) assessment of the impact of ChatGPT on learning outcomes, including at least one aspect of KSA, (c) involvement of participants from

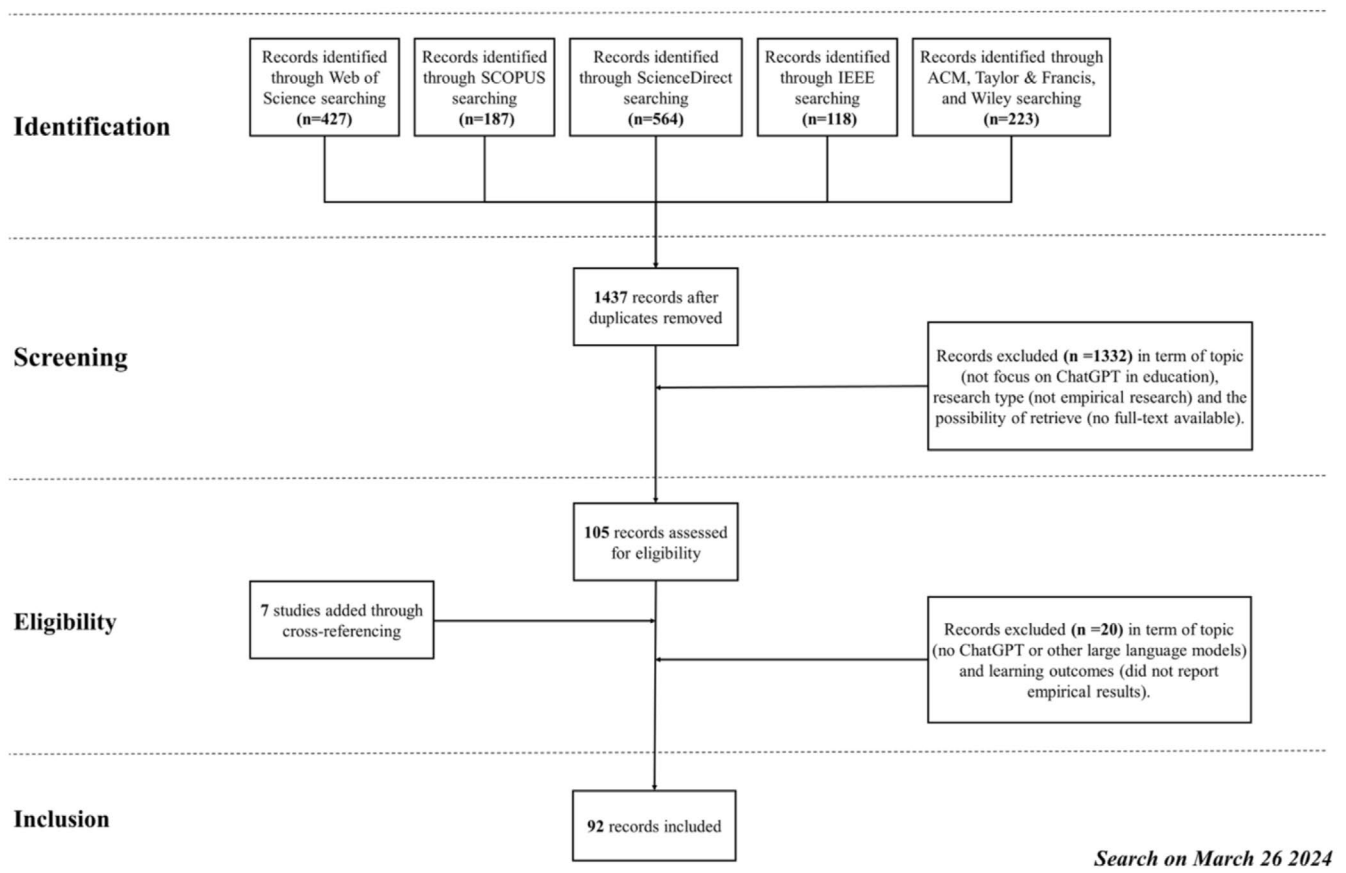


FIGURE 1 | The selection flowchart used based on PRISMA (Moher et al. 2009).

TABLE 1 | Search terms used in database searches.

ChatGPT search terms	Operator	Education search terms
'generative artificial intelligence' OR 'GAI' OR 'ChatGPT' OR 'GPT' OR 'large language model' OR 'LLM'	AND	'education' OR 'learning' OR 'teaching' OR 'class' OR 'course'

diverse age groups and academic disciplines, (d) use of empirical research methods and (e) publication in peer-reviewed journals or conference proceedings.

The exclusion criteria were applied to studies that: (a) did not focus on the educational application of ChatGPT, (b) failed to learning outcomes related to KSA, (c) were not empirical in nature or (d) were not published in English or were published before November 2022.

These inclusion and exclusion criteria were developed based on established guidelines for systematic reviews in educational technology research (Popay et al. 2006) and aligned with similar criteria used in recent systematic reviews of AI applications in education (Crompton and Burke 2023; Xu and Ouyang 2022). The criteria were specifically tailored to address our research questions regarding the distribution and types of ChatGPT applications in education and their influence on learning outcomes, ensuring a comprehensive yet focused review of the relevant literature.

3.4 | Quality Assessment Criteria

To ensure the methodological rigour and relevance of the included studies, we developed and applied a set of quality assessment criteria directly derived from our inclusion and exclusion criteria. Our quality assessment approach was informed by established frameworks used in previous systematic reviews of educational technology research (Petersen et al. 2015; Wohlin 2014), which emphasise the importance of evaluating studies based on their relevance, methodological quality and contribution to the research questions. Similar multi-criteria scoring systems have been successfully employed in systematic reviews of AI in education (Chen et al. 2020; Zawacki-Richter et al. 2019) and technology-enhanced learning (Hew and Cheung 2014). Two reviewers independently assessed each study using the following criteria:

1. Focus on ChatGPT in Education (0–2 points)
 - Does the study explicitly focus on the application of ChatGPT in educational settings?
2. Learning Outcome Assessment (0–2 points)
 - Does the study assess the impact of ChatGPT on learning outcomes related to knowledge, skills or attitudes?
3. Participant Diversity (0–2 points)
 - Does the study involve participants from diverse age groups and academic disciplines?
4. Empirical Research Methods (0–2 points)
 - Does the study employ appropriate empirical research methods?
5. Peer-Review Status (0–2 points)
 - Is the study published in a peer-reviewed journal or conference proceedings?

Each criterion was scored as 0 (not met), 1 (partially met) or 2 (fully met). The maximum possible score was 10 points. We established a 3-tier quality categorisation system based on the total scores: studies scoring 8 or above were considered high quality,

those scoring 6–7 were considered medium quality, and those scoring below 6 were considered low quality. These thresholds were determined through team consensus after reviewing the distribution of preliminary scores across the initial set of studies and considering the minimum acceptable quality standards for inclusion in our analysis. Only high and medium quality studies were included in the final analysis to ensure the robustness of our findings.

The three-tier quality categorisation system (high, medium, low) we employed aligns with quality assessment approaches used in previous educational technology systematic reviews (Bearman et al. 2012; Cheung and Slavin 2013), ensuring consistency with established practices in the field while being tailored to the specific context of ChatGPT in education research.

To ensure reliability of the quality assessment process, both reviewers independently assessed 20% of the articles, resulting in an inter-rater reliability score of 0.81 using Krippendorff's alpha (Krippendorff 2004), which exceeds the commonly accepted threshold of 0.80 for good reliability in content analysis (Neuendorf 2017). This reliability testing confirmed the consistency and dependability of our quality assessment approach before proceeding to independently code the remaining articles, with discrepancies resolved through consensus discussions.

3.5 | The Screening Process

The screening process entailed several procedures to ensure the rigour and relevance of the articles included in the analysis. Initially, duplicate articles were removed to maintain the uniqueness of the dataset. This step was followed by a careful examination of the titles and abstracts, whereby articles were excluded based on predetermined inclusion and exclusion criteria. Subsequently, a thorough review of the full texts led to further exclusions, adhering to the same criteria. Additionally, the snowballing technique was employed to identify further relevant articles via Google Scholar. Finally, data extraction from the meticulously filtered articles was conducted using Endnote 21 software. For visual representation of this process, refer to Figure 1.

In the initial search, 1519 articles were identified. After removing 82 duplicates, 1437 records remained. Upon reviewing titles and abstracts, 1332 articles were excluded based on criteria outlined in Section of quality assessment criteria, leaving 105 records. The first author examined the selected articles to ascertain their suitability for this systematic review, while the second author independently reviewed approximately 30% of these articles to ensure reliability. The inter-rater agreement rate was 90%. Subsequently, the first author assessed the full texts of the articles to confirm that they met all inclusion criteria for the review. During this process, 20 articles were identified as low quality and excluded from further analysis. Additionally, seven new articles were identified through snowball sampling and included in the review. As a result, a total of 92 articles were ultimately included in the systematic review (see Appendix S1 for the complete list of included studies with bibliographic details). This screening process followed the methodology outlined in

Ali et al. (2023), ensuring a systematic and transparent approach to study selection.

3.6 | General System Theory (GST)

This systematic review employs the GST as a framework to analyse the applications and impacts of ChatGPT in education. GST offers a holistic view of complex systems, considering them as interconnected elements that dynamically interact to fulfil a common purpose (Bertalanffy 1968; Xu and Ouyang 2022). Applying GST to the examination of ChatGPT in education facilitates an exploration of the interplay among five key elements:

- *Subject*: This element focuses on the main actors within the system, encompassing educators and learners, along with their roles, perceptions, experiences and characteristics. The analysis considers educational level (primary, secondary, higher education or instructors) and learning outcomes (knowledge, skills or attitudes). These dimensions are crucial for understanding ChatGPT's varying impacts across different educational stages and roles. Learning outcomes were categorised following the established KSA framework aligned with Bloom's taxonomy (Bloom et al. 1956; Krathwohl 2002) and widely used in educational assessment (Anderson and Krathwohl 2001). Specifically: Knowledge outcomes were operationalised as measurable improvements in factual understanding, conceptual comprehension, information retention and recall ability. Examples include enhanced subject matter knowledge, improved academic performance on tests, better concept understanding and increased information recall (Lee et al. 2024; Luo et al. 2025). Skills outcomes were defined as measurable improvements in applying knowledge through cognitive, technical or hands-on abilities. These included critical thinking, problem-solving, communication, writing proficiency, programming capability, data analysis and collaboration skills (Hu 2024; Lee, Cheng, et al. 2023; Weng et al. 2022). Attitudinal outcomes encompassed measurable changes in learners' feelings, values, motivations and engagement towards learning. These included increased learning motivation, improved self-efficacy, greater engagement with educational content, reduced anxiety, enhanced confidence and positive perceptions towards subject matter (De Loof et al. 2022; Lee, Lin, et al. 2023). Studies were coded to reflect all applicable outcome types based on the explicit measurements, assessments or observations reported in the research methodology and results. Many studies addressed multiple outcome types simultaneously, recognising the multidimensional nature of learning (Chen et al. 2025; Yeung et al. 2025).
- *Information*: This element pertains to the content, data or knowledge exchanged within the system, including learning materials, curricula and outcomes. It is examined through subjects or academic disciplines (ranging from language and STEM subjects to management and multidisciplinary studies) and sample sizes (small, medium or large scale). This analysis helps identify ChatGPT's effectiveness across various domains and the generalisability of findings. STEM subjects in this review were operationalised according to the widely accepted categorisation in educational research (Bybee 2013; English 2016) and include disciplines focused on Science (e.g., physics, chemistry, biology), Technology (e.g., information technology, informatics, programming), Engineering (e.g., mechanical, electrical, civil engineering) and Mathematics (e.g., algebra, calculus, statistics). The STEM categorisation reflects the integration of these disciplines in modern educational approaches, acknowledging their interconnectedness in addressing complex real-world problems (Kelley and Knowles 2016). Other subject categories were defined as follows: Language learning (including studies focused on language acquisition, literacy and linguistic skills); Humanities and Society (including history, philosophy, sociology, psychology and cultural studies); Medicine (studies focused on medical education and healthcare training); Business education (including management, economics, entrepreneurship and finance); and Multidisciplinary studies (research that spans multiple disciplines or doesn't fit clearly into the other categories).
- *Medium*: This element represents the channels, tools or platforms for information conveyance or interaction within the system, including web interfaces and Application Programming Interface (API) integration. By exploring ChatGPT's medium of delivery, such as web and API, the review assesses how accessibility and usability affect its adoption and effectiveness in educational settings.
- *Environment*: This element considers external factors, conditions or contexts that influence the system, including learning settings, institutional policies and sociocultural factors. The environmental context is analysed through various educational environments:
 - *Face-to-face environment*: Traditional classroom settings where instructors and learners interact in person, characterised by direct human-to-human communication and immediate feedback. These environments typically follow structured curricula within formal educational institutions.
 - *Experimental environment*: Hands-on, practice-oriented settings that prioritise active learning through direct engagement with materials, tools or simulations. These environments emphasise learning by doing, allowing students to apply theoretical knowledge to practical situations, and often involve laboratories, workshops or field-based activities where ChatGPT serves as a supportive tool.
 - *Informal learning environment*: Settings outside traditional classroom structures where learning occurs without explicit educational objectives or formal assessment. These include museums, libraries, science centres, online forums and self-directed learning contexts where ChatGPT functions as an on-demand resource for knowledge construction and exploration.
 - *Web-based environment*: Digital learning spaces that rely on internet-based technologies for content delivery and interaction. These range from structured online courses and learning management systems to virtual classrooms and digital collaboration platforms where ChatGPT is integrated as a learning support tool.

This investigation sheds light on how ChatGPT is integrated into these diverse instructional settings and how these environments impact its effectiveness in supporting learning outcomes.

- *Technology (Pedagogical Applications of Technology)*: While the original GST framework focuses on the specific tools, techniques or innovations employed within the system, our study emphasises the pedagogical applications of these technological artefacts in educational contexts. This adaptation is necessitated by the nature of ChatGPT as an educational tool, where its impact is primarily realised through its pedagogical implementation rather than its technical specifications. Therefore, this element examines how ChatGPT is applied in educational settings to support and enhance learning processes. The pedagogical application dimension encompasses automated assessment, personalised learning, timely feedback and content generation. This analysis provides insight into how ChatGPT is utilised to support and enhance educational processes, reflecting the intersection of technology and pedagogy in the context of AI-enhanced education.

By mapping collected data to these five GST elements and their respective dimensions, the review offers a comprehensive understanding of ChatGPT's applications and impact in education. The GST framework facilitates a systematic analysis of the interactions between ChatGPT and various components of the educational ecosystem, informing future research and practice in this rapidly evolving field. The selection of variables and dimensions within each element captures key factors influencing ChatGPT's effectiveness in diverse educational settings, laying a solid foundation for evaluating its potential and limitations.

3.7 | Data Extraction and Synthesis

In this study, the content analysis method, as outlined by Cohen et al. (2002) and Zupic and Čater (2014), was employed to categorise 92 articles on ChatGPT's application in education, addressing the research questions. GST, a coding framework for educational system elements, was crafted to methodically review articles on ChatGPT in the educational sector (refer to Table 2). This framework encompassed various dimensions: the subject (detailing educational levels and learning outcomes), information (including subjects and sample sizes), medium (ChatGPT's medium of delivery), environment (educational contexts) and technology (ChatGPT techniques utilised).

Two raters coded the 92 articles. An article could receive multiple codes within a single dimension. Initially, to ensure coding reliability, 20% of the articles were independently coded by the raters, resulting in a Krippendorff (2004) alpha reliability score of 0.87. Following this reliability confirmation, the remaining articles were independently coded by the raters. Discrepancies in coding were resolved through consensus. Detailed examples and explanations are provided to elucidate how the coding results reflect the data reviewed, in line with Graneheim and Lundman (2004) and (Xu and Ouyang 2022).

The extracted data were synthesised using a narrative approach, organising the findings by the key elements of the GST framework and their respective dimensions. Descriptive

TABLE 2 | The coding scheme.

Element	Dimension	Type
Subject	Educational level	1. Primary education (6–12) 2. Secondary education (12–18) 3. Higher education (> 18)
	Learning outcomes	1. Knowledge 2. Skills 3. Attitudes
Information	Learning content	1. Language learning 2. STEM subjects 3. Humanities and Society 4. Medicine 5. Business education 6. Multidisciplinary studies
	Sample size	1. Small scale (≤ 50) 2. Medium scale (51–300) 3. Large scale (> 300)
Medium	ChatGPT medium	1. ChatGPT web 2. ChatGPT API
Environment	Educational context	1. Face-to-face environment 2. Experimental environment 3. Informal learning environment 4. Web-based environment
Technology	ChatGPT technique	1. Automated assessment 2. Personalised learning 3. Timely feedback 4. Content generation

statistics summarised the distribution of ChatGPT applications across different categories, while qualitative themes were identified to capture key patterns and relationships among the GST elements. The synthesis aimed to provide a comprehensive and nuanced understanding of ChatGPT's applications and impact in education, considering the complex interactions among subjects, information, mediums, environments and technologies.

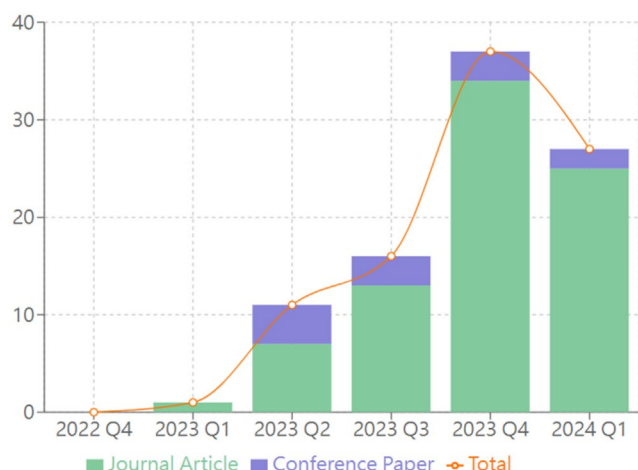
4 | Results

4.1 | Distribution and Types of ChatGPT Applications in Relation to the GST Framework

This section explores the distribution and types of ChatGPT applications within the five key elements of the GST framework: subject, information, medium, environment and technology. Analysing 92 studies from this systematic review provides a comprehensive view of ChatGPT's implementation in diverse educational settings. To provide a clear overview of the temporal distribution of the selected studies, Table 3 presents the number of papers published quarterly from Q4 2022 to Q1 2024. This

TABLE 3 | Quarterly publication trends.

Quarter	Journal articles	Conference papers	Total
2022 Q4	0	0	0
2023 Q1	1	0	1
2023 Q2	7	4	11
2023 Q3	13	3	16
2023 Q4	34	3	37
2024 Q1	25	2	27

**FIGURE 2** | Month distribution in articles on ChatGPT application.

distribution reflects the rapid growth of research on ChatGPT in education following its release in late 2022.

As shown in Table 3, there is a clear trend of increasing research output over time, with a notable surge in publications from Q3 2023 onwards. The data reveals a significant jump from 16 publications in Q3 2023 to 37 publications in Q4 2023, marking the most substantial increase in the observed period. Journal articles consistently outnumber conference papers across all quarters, indicating a preference for journal publications in this rapidly evolving field.

The research trend demonstrates a marked increase in publications from Q4 2022, where no publications were recorded, to Q1 2024, which saw 27 publications. This trend underscores the growing interest and rapid development of research on ChatGPT applications in education. The visual representation in Figure 2 further illustrates this quarterly distribution and the overall upward trajectory of research output in this domain.

4.2 | Subject

In the subject element, the analysis of educational levels reveals that 69.7% of studies focus on higher education, 13.5% on secondary education and 3.4% on primary education (Figure 3). The high concentration of research in tertiary education indicates that ChatGPT is more widely explored and applied in post-secondary

settings, possibly due to the more advanced nature of the content and the greater autonomy of learners at this level.

Analysis of learning outcomes reveals that knowledge construction is the most frequently investigated dimension, with 63 mentions across the included studies. Skills development follows closely with 59 mentions, while attitudinal outcomes receive comparatively less attention, with 20 mentions (Figure 4). This suggests that ChatGPT is more commonly employed to support the acquisition of knowledge and skills, while its impact on attitudes is less frequently examined.

4.3 | Information

Within the information element, STEM subjects constitute the most researched category, with 45.6% of studies focusing on this domain. Humanities and Social Sciences follow with 15.6%, Language Learning (12.2%), Multidisciplinary Studies (17.8%), Business Education (5.6%) and Medicine (2.2%) (Figure 5). This distribution highlights ChatGPT's versatility in supporting learning across a wide range of academic disciplines, particularly in STEM fields where its ability to provide explanations and engage in problem-solving is highly valued.

Regarding sample size, 40.4% of studies involve small samples (≤ 50 participants), 43.8% involve medium-sized samples (51–300 participants), and 14.6% involve large samples (> 300 participants) (Figure 6). The prevalence of small and medium-sized samples suggests that many studies on ChatGPT in education are still exploratory, and more large-scale investigations are needed to establish the generalisability of the findings.

4.4 | Medium

All 92 studies specified the ChatGPT medium used, with 80.4% utilising a web-based version of ChatGPT and 19.6% employing a ChatGPT API (Figure 7). The dominance of web-based ChatGPT underscores its accessibility and ease of use, making it a popular choice for educational research and practice.

4.5 | Environment

Among the 71 studies specifying the educational context, 57.7% focus on web-based environments, 23.9% on face-to-face environments, 16.9% on experimental environments and 1.4% on informal learning environments (Figure 8). The prominence of web-based environments reflects the growing importance of online learning and ChatGPT's potential to enhance these experiences. The limited representation of informal learning environments suggests an opportunity for future research to explore ChatGPT's application in less-structured educational settings.

4.6 | Technology

Timely feedback emerges as the most frequently employed ChatGPT technique, with 55 mentions across the included studies. Content generation follows with 20 mentions, while

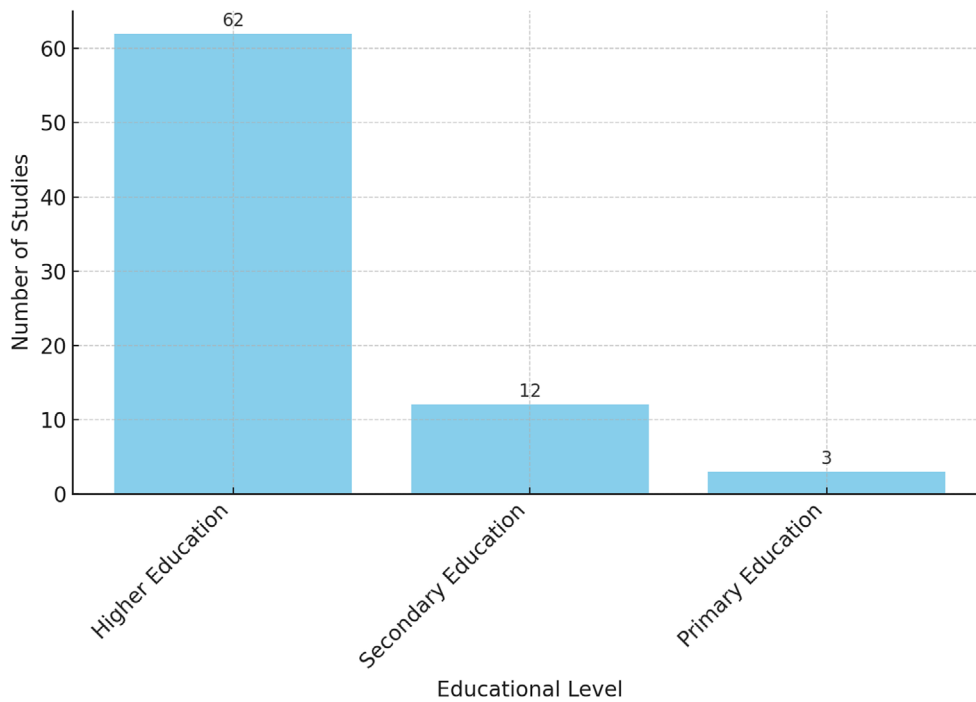


FIGURE 3 | The distribution of educational level.

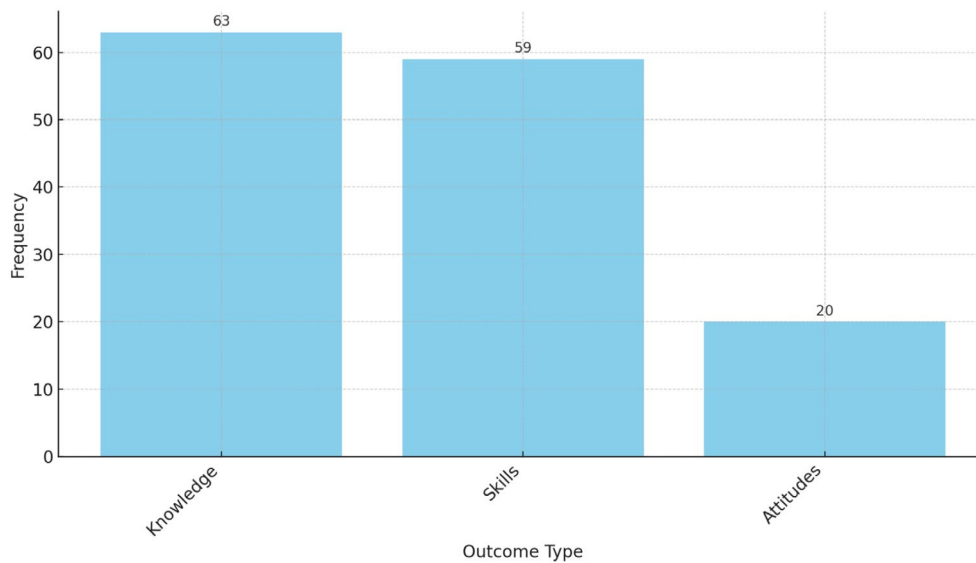


FIGURE 4 | The distribution of learning outcomes.

personalised learning and automated assessment are mentioned 15 and 12 times, respectively (Figure 9). The high frequency of timely feedback highlights ChatGPT's potential to provide immediate and targeted support to learners, with content generation, personalised learning and automated assessment representing other promising applications in education.

4.7 | Influence of Subject, Information, Medium and Environment Elements on the Technology Element

This section explores how the subject, information, medium and environment elements influence the technology element,

specifically the ChatGPT techniques employed in educational contexts.

4.7.1 | Subject and Technology

Figure 10 shows the deployment of ChatGPT methodologies across different educational levels. In higher education, the predominant applications include timely feedback ($N=38$) and content generation ($N=12$). In secondary education, there is a focus on timely feedback ($N=11$), with lesser emphasis on automated assessment ($N=2$) and personalised learning ($N=2$); content generation is absent in this tier. Primary education is characterised by minimal yet significant

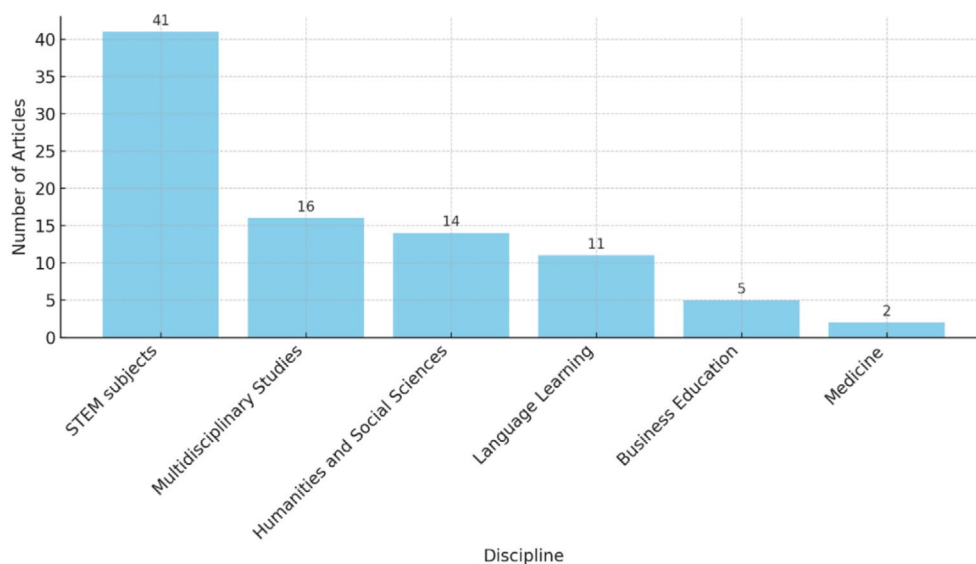


FIGURE 5 | The distribution of learning content.

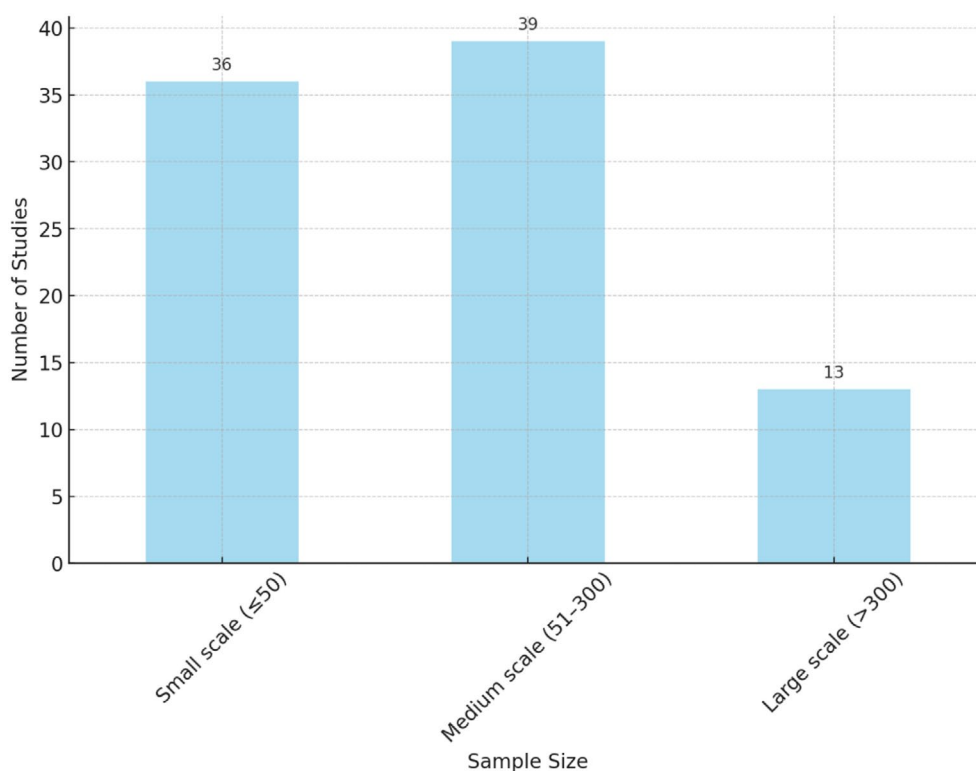


FIGURE 6 | The distribution of sample size.

mentions of timely feedback ($N=1$) and content generation ($N=2$).

Figure 11 delineates the correlation between various ChatGPT methodologies and their impacts on learning outcomes. For knowledge construction, timely feedback ($N=34$) and content generation ($N=16$) emerge as predominant methods. For skills development, a similar pattern is observed, with timely feedback ($N=34$) and content generation ($N=14$) being the primary techniques. Techniques aimed at influencing attitudinal outcomes

appear less frequent, with timely feedback ($N=11$) and personalised learning ($N=6$) being utilised.

4.7.2 | Information and Technology

Figure 12 showcases the application of ChatGPT techniques across different learning content areas. In STEM subjects, timely feedback ($N=28$) and personalised learning ($N=7$) are the most prevalent techniques. Humanities and social sciences

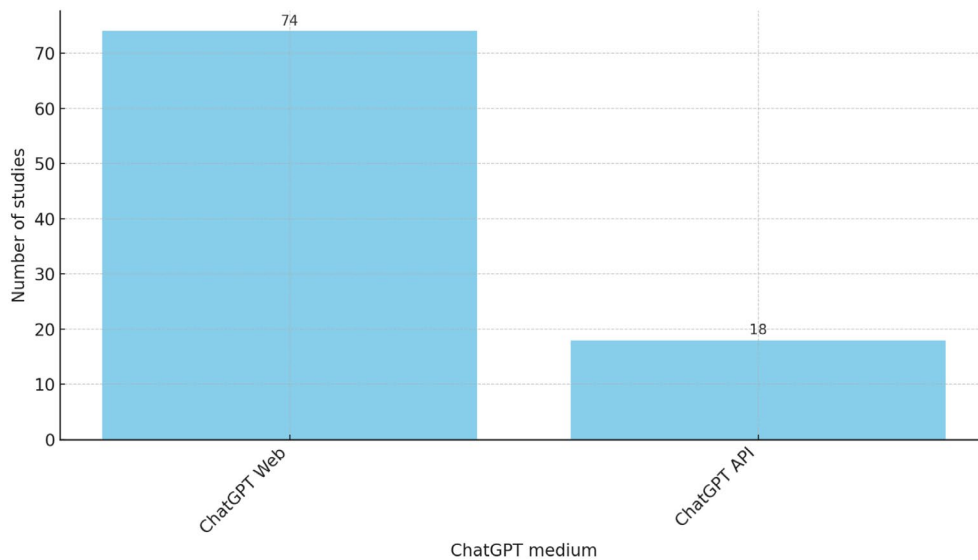


FIGURE 7 | The distribution of ChatGPT medium.

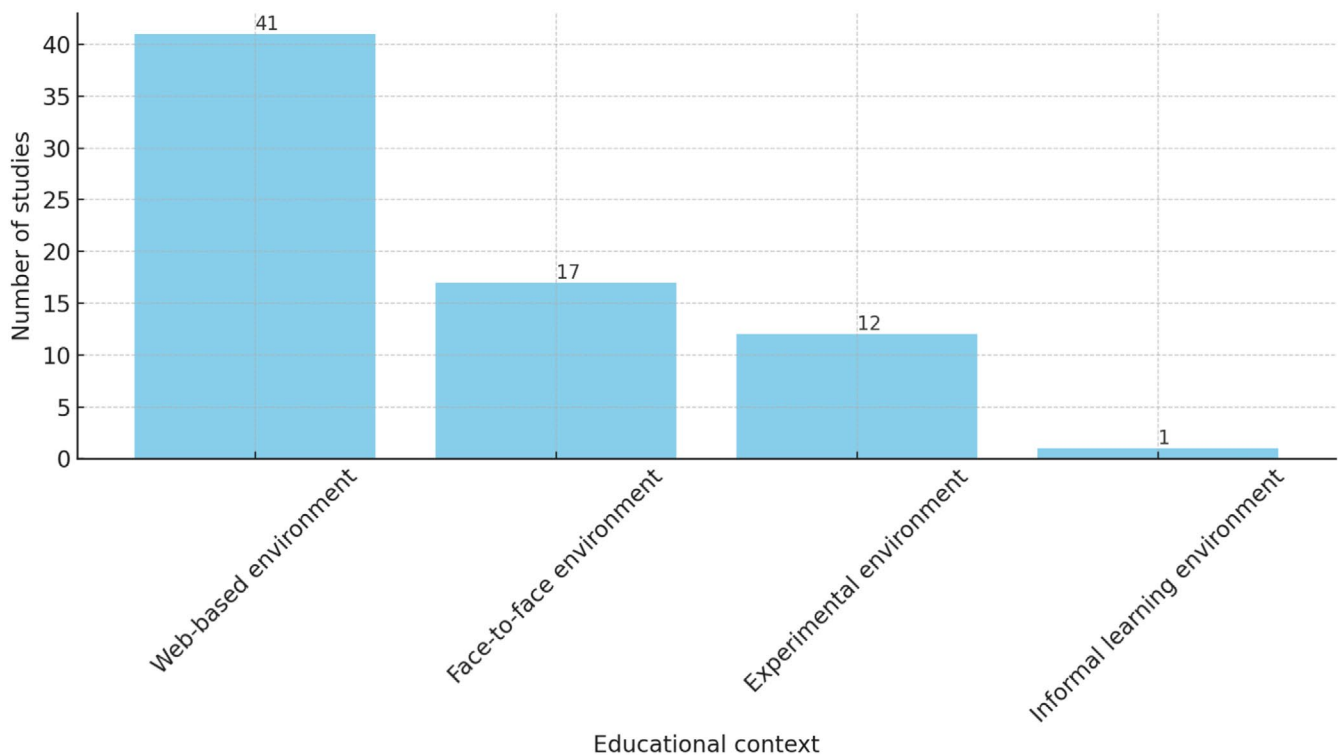


FIGURE 8 | The distribution of educational Context.

primarily utilise content generation ($N=7$) and timely feedback ($N=6$). Language learning employs timely feedback ($N=8$), content generation ($N=2$), personalised learning ($N=2$) and automated assessment ($N=2$). Multidisciplinary studies leverage timely feedback ($N=7$) and content generation ($N=5$). Business education mainly uses timely feedback ($N=4$) and personalised learning ($N=1$), while medicine only reports the use of timely feedback ($N=1$) and personalised learning ($N=1$).

Figure 13 illustrates the relationship between ChatGPT techniques and sample sizes in the reviewed studies. For

small-scale studies (≤ 50), timely feedback ($N=20$) and content generation ($N=11$) are the most common techniques. Medium-scale studies (51–300) primarily employ timely feedback ($N=28$) and content generation ($N=6$). Large-scale studies (> 300) mainly utilise timely feedback ($N=6$) and personalised learning ($N=4$).

4.7.3 | Medium and Technology

Figure 14 depicts the application of ChatGPT techniques under different ChatGPT mediums. For the ChatGPT web medium,

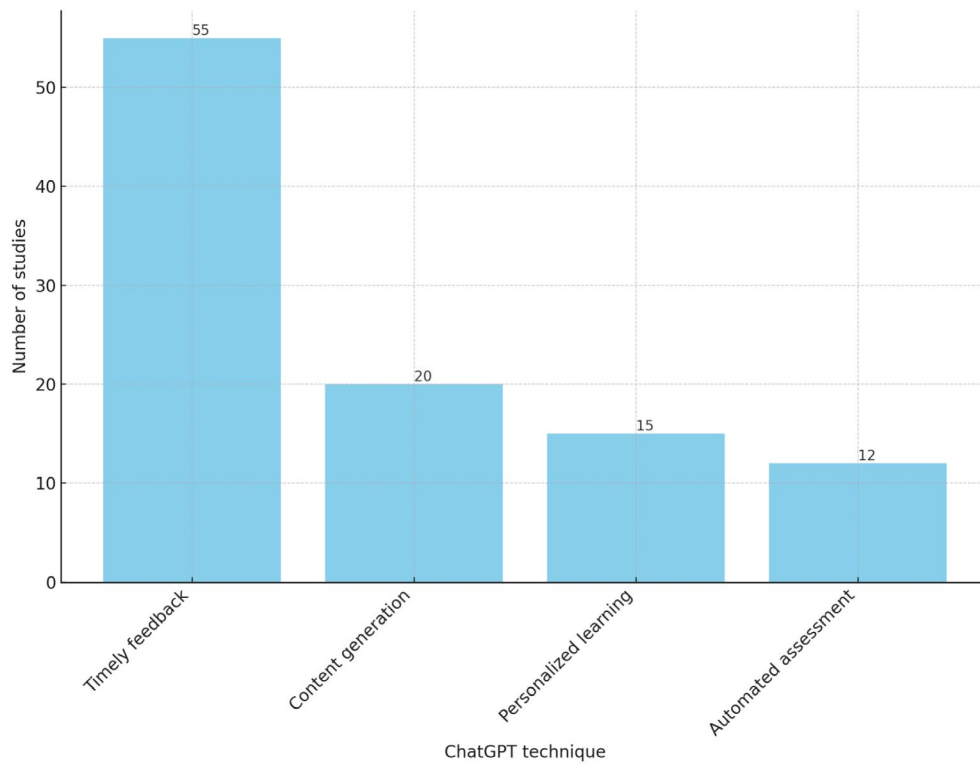


FIGURE 9 | The distribution of ChatGPT technique.

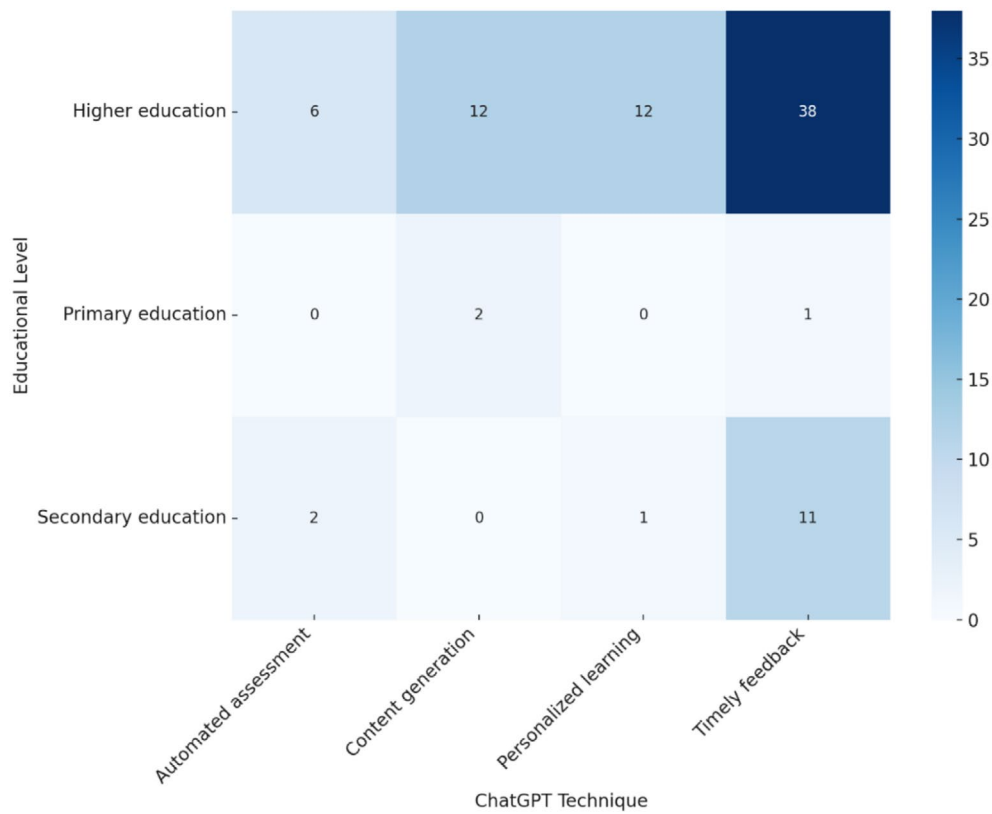


FIGURE 10 | The distribution of ChatGPT applications under educational level.

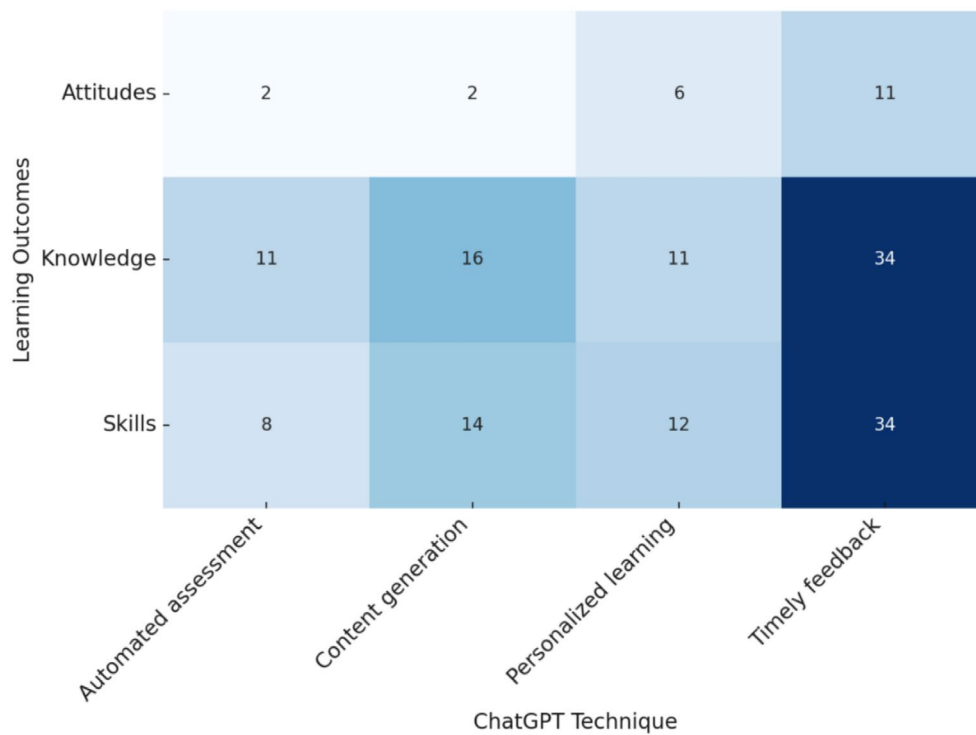


FIGURE 11 | The distribution of ChatGPT applications under learning outcomes.

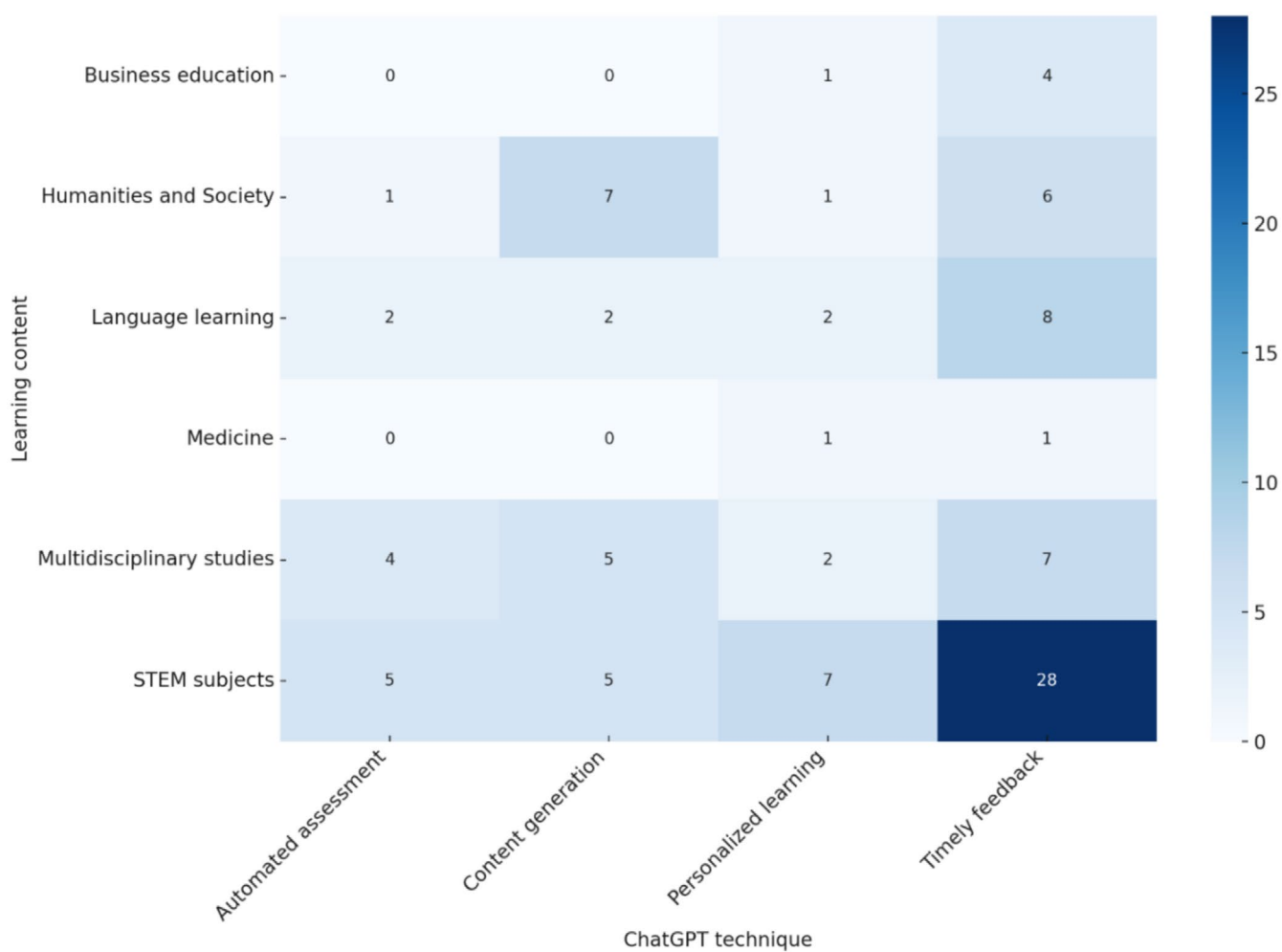


FIGURE 12 | The distribution of ChatGPT applications under learning content.

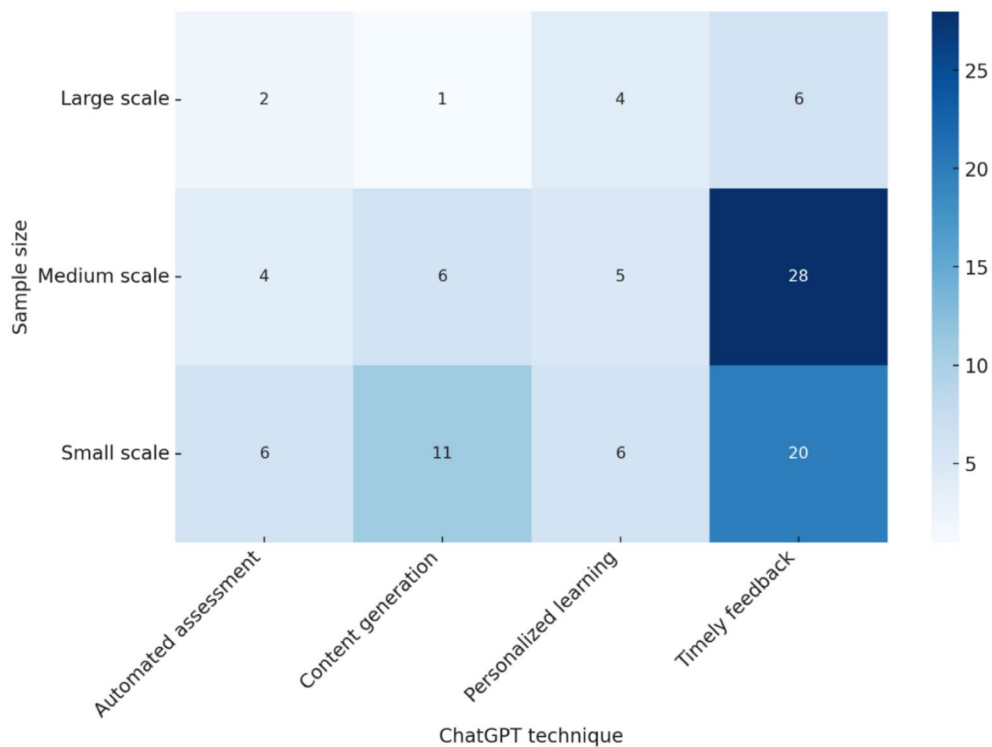


FIGURE 13 | The distribution of ChatGPT applications under sample size.

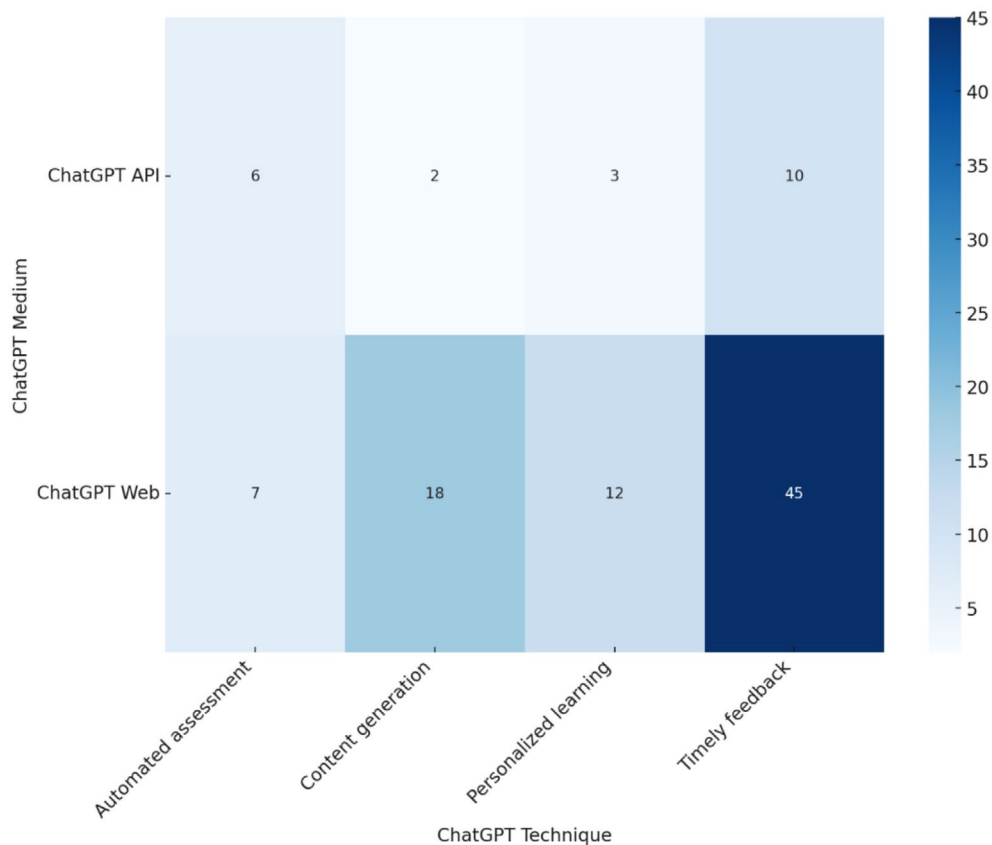


FIGURE 14 | The distribution of ChatGPT applications under its medium.

timely feedback ($N=45$) and content generation ($N=18$) are the most frequently used techniques. The ChatGPT API medium primarily employs timely feedback ($N=10$) and automated assessment ($N=6$).

4.7.4 | Environment and Technology

Figure 15 presents the distribution of ChatGPT techniques across various educational contexts. In the web-based environment, timely feedback ($N=23$) and content generation ($N=11$) are the most prevalent techniques. Face-to-face environments mainly utilise timely feedback ($N=11$), content generation ($N=3$) and automated assessment ($N=3$). Experimental environments primarily employ timely feedback ($N=8$) and personalised learning ($N=3$). The informal learning environment only reports the use of content generation ($N=1$).

In summary, the analysis of the interactions between the subject, information, medium, environment and technology elements reveals patterns in the application of ChatGPT techniques in educational contexts. Timely feedback emerges as the most frequently employed technique across all dimensions, followed by content generation, personalised learning and automated assessment. The prevalence of these techniques varies depending on the educational levels, learning outcomes, content areas, sample sizes, mediums and educational contexts. These findings provide valuable

insights into how the various elements of the GST framework influence the selection and implementation of ChatGPT techniques in education, informing future research and practice in this rapidly evolving field.

5 | Discussion

While our systematic review highlights the potential benefits of ChatGPT in education, it is important to address potential counterarguments to its widespread adoption. One significant concern is the potential for ChatGPT to hinder critical thinking skills and promote intellectual laziness among students (Lee et al. 2024; Sallam et al. 2023). Critics argue that easy access to AI-generated responses might discourage students from engaging deeply with course material, leading to superficial learning. However, our analysis suggests that when properly integrated into educational practices, ChatGPT can actually enhance critical thinking skills rather than diminish them. For instance, Lee et al. (2024) demonstrated that using ChatGPT as a tool for dialogue and debate stimulated students to question, analyse and evaluate information more critically. Moreover, Nguyen et al. (2024) found that students who used ChatGPT as a brainstorming tool showed improved ability to synthesise complex ideas and construct well-reasoned arguments. These findings indicate that the key to harnessing ChatGPT's potential lies in its thoughtful implementation as a complement to, rather than

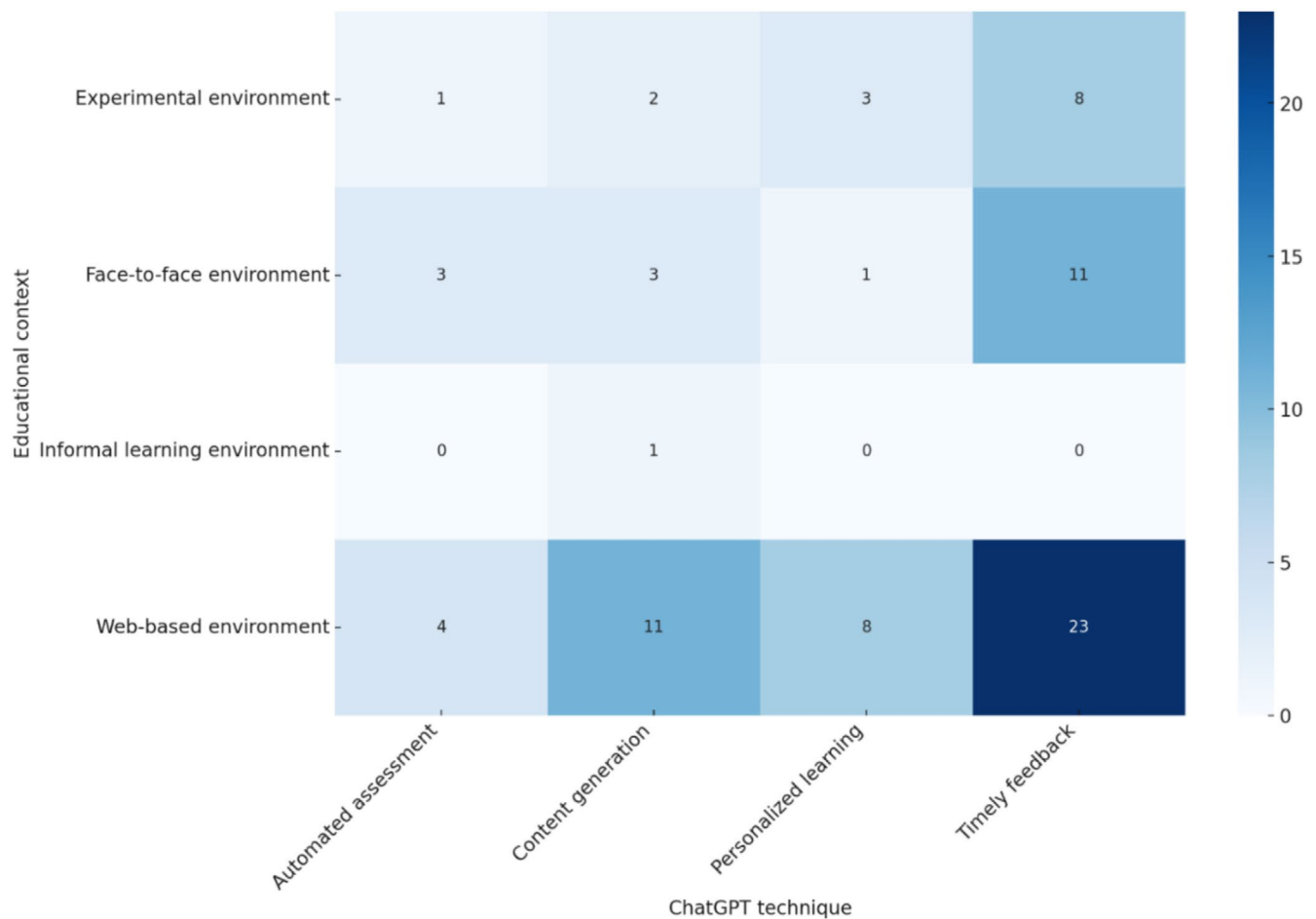


FIGURE 15 | The distribution of ChatGPT applications under educational context.

a replacement for, traditional learning methods. By encouraging students to critically engage with AI-generated content, educators can foster a more nuanced understanding of both the subject matter and the capabilities and limitations of AI technologies.

5.1 | The Distribution and Types of ChatGPT Applications in Relation to the GST Framework

The systematic review of ChatGPT applications in education, analysed through the GST framework, illuminates the current research landscape and outcomes in this rapidly evolving field. The high concentration of research in higher education settings (69.7%) underscores ChatGPT's potential to enhance student learning experiences in post-secondary contexts, likely due to the advanced content and greater learner autonomy (Lee et al. 2024). This focus has led to improved personalised learning experiences, increased student engagement and enhanced knowledge retention. However, the scarcity of research in primary (3.4%) and secondary education (13.5%) underscores the need for studies on age-appropriate interventions that address younger students' unique needs and learning styles (Vázquez-Cano et al. 2023). Future studies should explore how ChatGPT can support learning across all educational levels to promote more effective learning outcomes, potentially leading to improved educational experiences for students of all ages.

The distribution of research highlights a strong emphasis on knowledge (63 mentions) and skills construction (59 mentions), with less attention to attitudinal outcomes (20 mentions). While focusing on cognitive and practical aspects aligns with ChatGPT's ability to provide personalised responses and engage in interactive discussions (Wu et al. 2024), it has resulted in outcomes such as improved subject-specific knowledge, enhanced problem-solving skills, and more effective information retention. The limited attention to attitudinal outcomes indicates a gap in understanding how ChatGPT influences learners' dispositions and behaviours towards learning (Kohnke et al. 2023). The few studies addressing this aspect suggest potential outcomes like increased learning motivation and improved self-efficacy. Future studies should examine the affective dimensions of learning in relation to ChatGPT interventions, as fostering positive attitudes and motivation is crucial for long-term academic success.

ChatGPT's versatility in supporting learning across various disciplines is evident, with a notable focus on STEM subjects (45.6%). This reflects ChatGPT's utility in providing timely feedback and engaging in problem-solving discussions, valuable in STEM where conceptual understanding and application are critical (Kazemitabaar et al. 2024). Outcomes in STEM education include improved conceptual understanding, enhanced problem-solving skills, and increased proficiency in complex scientific and mathematical concepts. Specifically, studies have documented significant improvements in technical skills such as programming capabilities through step-by-step guidance and debugging assistance (Kazemitabaar et al. 2024), data visualisation and statistical interpretation skills through interactive problem-solving sessions (Alfirević et al. 2024; Huang et al. 2024), and scientific reasoning skills that bridge theoretical knowledge with practical applications (Allen et al. 2024;

Valeri et al. 2025). Students using ChatGPT in STEM contexts also demonstrated enhanced technical writing proficiency and improved abilities to articulate complex information (Woolever and Wettstein 2024), as well as developed collaborative problem-solving competencies through simulated discussions and project management support (Jony and Hamim 2024). However, the underrepresentation of Humanities and Social Sciences (15.6%) and Language Learning (12.2%) points to the need for further research on ChatGPT's application in fostering critical thinking, creative expression and cultural understanding. The limited studies in these areas suggest potential outcomes such as improved writing skills, enhanced language proficiency and more nuanced understanding of cultural contexts. In language learning contexts, specific skill improvements included enhanced grammatical accuracy, expanded vocabulary acquisition and improved reading comprehension (Dakakni and Safa 2023; Vázquez-Cano et al. 2023), while in humanities, students demonstrated strengthened argumentation skills, enhanced ability to synthesise complex ideas and improved critical analysis of texts (Nguyen et al. 2024). Exploring its potential across a broader range of disciplines can lead to innovative pedagogical approaches that enhance learning experiences in diverse fields.

The predominance of STEM disciplines (45.6%) in ChatGPT education research is particularly noteworthy when examined in relation to learning outcomes. Analysis reveals that STEM-focused studies frequently address multiple learning outcome dimensions simultaneously, with 68% of STEM studies measuring both knowledge construction and skills development, and 27% incorporating all three dimensions (knowledge, skills and attitudes). This multidimensional approach in STEM research reflects the complex nature of STEM education, which often aims to develop not only subject-specific knowledge but also practical problem-solving abilities and positive dispositions towards scientific inquiry (Fairhurst et al. 2023; Lee, Cheng, et al. 2023; Wu et al. 2025). For instance, Lee et al. (2024) examined how ChatGPT-enhanced blended learning affects students' self-regulated learning, higher-order thinking skills and knowledge construction in STEM education. Similarly, Wu et al. (2024) demonstrated ChatGPT's capacity to promote both self-regulation (attitude) and knowledge construction in technical subjects. This multifaceted approach to outcomes assessment in STEM aligns with contemporary educational frameworks that recognise the interconnected nature of cognitive, psychomotor and affective domains in effective learning (Anderson and Krathwohl 2001; Krathwohl 2002).

The prevalence of studies using small (40.4%) and medium-sized samples (43.8%) indicates that research on ChatGPT in education is still exploratory. While these studies offer valuable insights and suggest positive outcomes in terms of improved learning experiences and knowledge acquisition, the scarcity of large-scale investigations (14.6%) highlights the need for comprehensive research to establish generalisability (Meyer et al. 2024). Large-scale studies could provide more robust evidence of ChatGPT's impact on standardised test scores, long-term knowledge retention and overall academic performance. Future studies should prioritise large-scale, multi-institutional investigations to provide robust evidence of ChatGPT's impact on learning outcomes and adaptability to diverse educational contexts.

The dominance of web-based ChatGPT applications (80.4%) underscores the significance of online learning and the seamless integration of AI technologies into educational platforms. This trend has resulted in enhanced virtual learning experiences, improved accessibility to educational support, and increased flexibility in learning schedules. However, the limited exploration of ChatGPT API (19.6%) suggests opportunities for more advanced, customised applications in education (Dehbozorgi and Kunuku 2023). Future research should investigate the development of ChatGPT-based tools leveraging the API to create tailored learning experiences and support specific educational goals, potentially leading to more personalised and effective learning outcomes.

In the environment element, the emphasis on web-based environments (57.7%) reflects the growing importance of online learning and ChatGPT's potential to enhance virtual experiences. This focus has led to improved remote learning outcomes, increased student participation in online discussions and enhanced digital literacy skills. Web-based ChatGPT applications have demonstrated particular effectiveness in developing specific skills, including self-regulated learning strategies and metacognitive abilities (Lee et al. 2024; Wu et al. 2024), digital communication competencies and virtual collaboration skills (Huang et al. 2025), and information literacy skills such as evaluating source credibility and synthesising information from multiple sources (Chen et al. 2025). In face-to-face environments, ChatGPT integration has shown success in developing different skill sets, including enhanced in-person collaborative problem-solving abilities, improved classroom discussion participation skills, and strengthened peer feedback competencies (Wu et al. 2025). Experimental environments utilising ChatGPT have particularly excelled at developing hands-on technical skills, laboratory report writing proficiency and experimental design abilities (Li et al. 2024), while the limited studies in informal learning settings suggest potential for developing self-directed learning skills and inquiry-based competencies (Tupper et al. 2024). However, the limited representation of face-to-face (23.9%), experimental (16.9%) and informal learning environments (1.4%) indicates a need for broader research on ChatGPT's integration into various instructional settings. Understanding the unique affordances and constraints of different environments is crucial for optimising ChatGPT's educational impact and achieving diverse learning outcomes across various contexts.

The technology element showcases diverse applications of ChatGPT techniques in education, with timely feedback, content generation, personalised learning and automated assessment being the most utilised. The high frequency of timely feedback highlights ChatGPT's capacity to provide immediate, targeted support, underscoring the value of real-time feedback in enhancing learning and engagement (Rasul et al. 2023). This has resulted in improved learning efficiency, faster error correction and enhanced student confidence. The focus on content generation and personalised learning points to ChatGPT's potential to reduce educators' workload and cater to individual learners' needs, while automated assessment reflects its capacity to streamline grading processes (Pinto et al. 2023). These applications have led to outcomes such as more diverse learning materials, increased adaptation to individual learning styles, and more

consistent and timely assessment feedback. Future research should design interventions leveraging these techniques to optimise learning outcomes and support diverse learner needs.

5.2 | The Influence of Subject, Information, Medium and Environment Elements on the Technology Element

Analysing interactions among the GST elements reveals significant patterns in applying ChatGPT techniques in educational contexts. For the subject element, timely feedback is the most frequently used technique for both learners and instructors, followed by content generation. This pattern suggests that ChatGPT's ability to provide immediate, targeted feedback is highly valued, aligning with formative assessment principles and scaffolding in educational theories (Black and Wiliam 2009; Vygotsky and Cole 1978). Outcomes include improved learning efficiency and enhanced understanding of complex concepts. The prominence of content generation and personalised learning for learners highlights ChatGPT's potential to cater to individual needs, reflecting the emphasis on student-centred approaches (Chiu 2024; Wu et al. 2024). For instructors, the focus on content generation and automated assessment suggests ChatGPT can aid in creating high-quality materials and streamlining grading, potentially reducing workload and allowing more time for student engagement (Pinto et al. 2023; Rasul et al. 2023).

The distribution of ChatGPT techniques across educational levels shows a predominance of timely feedback and content generation in higher education, with limited application in secondary and primary education. This indicates a need for research on age-appropriate ChatGPT interventions that address younger students' developmental needs (Vázquez-Cano et al. 2023). Outcomes in higher education include enhanced critical thinking and research skills, while potential benefits for younger learners remain largely unexplored. The absence of content generation in secondary education and minimal use in primary education highlight the necessity to adapt ChatGPT for younger learners, considering factors like cognitive development and curriculum requirements.

Regarding learning outcomes, there is a strong emphasis on timely feedback and content generation for knowledge construction and skills development. This aligns with ChatGPT's ability to provide personalised explanations and targeted feedback (Wu et al. 2024), resulting in improved subject understanding and problem-solving abilities. The lower frequency of techniques aimed at influencing attitudinal outcomes highlights a gap in understanding how ChatGPT can shape learners' dispositions and behaviours towards learning (Kohnke et al. 2023). Future studies should explore designing ChatGPT interventions that foster positive attitudes and motivation, crucial for long-term success.

Within the information element, ChatGPT techniques vary across disciplines, with timely feedback being the most prevalent. The prominence of personalised learning in STEM subjects suggests ChatGPT is used to provide targeted support in these domains, where conceptual understanding is critical (Kazemitabaar

et al. 2024). Outcomes include improved conceptual grasp and analytical skills. The emphasis on content generation in Humanities and Social Sciences highlights ChatGPT's potential to support engaging materials that foster critical thinking and creativity. Limited use in language learning, business and medicine underscores the need for research on discipline-specific applications, considering each field's unique requirements.

Sample size analysis reveals a predominance of timely feedback and content generation in small and medium-scale studies. The limited large-scale investigations highlight the need for comprehensive research to establish generalisability and scalability (Meyer et al. 2024). Future studies should prioritise large-scale, multi-institutional investigations to provide robust evidence of ChatGPT's impact across diverse contexts.

The medium element shows that timely feedback and content generation are most frequently used in web-based environments, while the ChatGPT API primarily employs timely feedback and automated assessment. This suggests web-based applications are versatile and adaptable to various educational needs, aligning with the importance of online learning (Dehbozorgi and Kunuku 2023). The limited use of content generation and personalised learning in the ChatGPT API highlights an opportunity to explore advanced, customised applications leveraging the API for tailored experiences.

Across the environment element, timely feedback is the most prevalent technique in web-based, face-to-face and experimental settings, while content generation is reported in informal learning environments. The prominence of timely feedback underscores its adaptability and effectiveness in various contexts, enhancing learner engagement and outcomes (Rasul et al. 2023). Limited representation of other techniques in face-to-face, experimental and informal environments highlights the need for research on integrating ChatGPT into diverse instructional settings, considering each context's unique affordances and constraints (Bower et al. 2024). Future studies should explore ChatGPT interventions that leverage the strengths of each environment, such as rich social interactions in face-to-face settings or self-directed learning opportunities in informal contexts.

The analysis of interactions among the GST elements underscores the importance of considering the unique characteristics of learners, learning objectives, content areas, mediums, environments and targeted outcomes when designing and implementing ChatGPT interventions. A holistic, context-sensitive approach is essential for maximising the effectiveness of ChatGPT techniques in diverse educational settings. Future research should examine how the interrelationships among GST elements influence the selection and effectiveness of ChatGPT techniques, identifying best practices for enhancing teaching and learning processes.

5.3 | Current Limitations in ChatGPT Research in Education and Future Research Directions

The systematic review reveals several limitations in current ChatGPT research in education and highlights areas for

future investigation. One notable limitation is the overemphasis on higher education settings (69.7%), with minimal research on primary (3.4%) and secondary education (13.5%). This focus hinders understanding of ChatGPT's adaptability and effectiveness in supporting younger learners with diverse developmental needs and learning styles. This imbalance results in a skewed understanding of its educational impacts, with comprehensive knowledge about higher education but limited insights into its potential benefits and challenges for younger learners.

To address this, future research should prioritise age-appropriate ChatGPT interventions across all educational levels. This includes exploring feasibility and effectiveness in primary and secondary education, considering cognitive development, digital literacy and curriculum requirements. Expanding the research scope will provide valuable insights into ChatGPT's versatility and potential to support learning outcomes across all age groups.

Furthermore, while this review synthesises international literature, the search was limited to English-language publications, and the distribution of the included studies may reflect a publication bias towards certain geographical regions (e.g., North America, Europe, East Asia). Consequently, perspectives from non-English speaking regions, particularly in the Global South, may be underrepresented, limiting the full international scope of the findings. Future research should therefore endeavour to incorporate a more linguistically and geographically balanced body of literature to enrich the global dialogue.

Another limitation is the predominance of small to medium-scale studies, with only 14.6% involving large samples. This reliance limits the generalisability of findings and understanding of ChatGPT's effectiveness and scalability across diverse demographics, levels and disciplines. Future studies should prioritise large-scale, multi-institutional research to provide robust evidence of ChatGPT's impact and adaptability (Meyer et al. 2024). This approach will yield more comprehensive and representative findings, informing evidence-based practices and policies.

The review also highlights a need for more research on ChatGPT's impact on attitudinal outcomes, as most studies focus on knowledge and skills construction. Examining the affective dimensions of learning related to ChatGPT interventions will contribute to a holistic understanding of its potential to shape learners' attitudes and behaviours, informing strategies to promote positive learning experiences (Cai et al. 2023; Song and Song 2023).

Furthermore, there is a lack of comprehensive frameworks and guidelines for effective ChatGPT integration in education. This limitation has led to fragmented approaches, potentially causing inconsistent outcomes. Future research should focus on developing evidence-based frameworks and guidelines, considering the interplay of GST elements (Su and Yang 2023). Establishing clear guidelines will optimise ChatGPT's potential to enhance teaching and learning processes, promoting effective and responsible integration.

In conclusion, addressing these limitations is essential for advancing ChatGPT research in education. Expanding research to include all educational levels, conducting large-scale studies, focusing on attitudinal outcomes and developing comprehensive frameworks will contribute to a more holistic understanding of ChatGPT's potential. This will provide a stronger foundation for leveraging ChatGPT to enhance learning outcomes and meet the evolving needs of 21st-century education.

5.4 | Global Perspectives and Contextual Implications

The findings of this systematic review, when viewed from an international perspective, offer crucial insights into global trends in educational technology. The pronounced concentration of research in higher education (69.7%) is not merely a data point but reflects a widespread pattern where universities often serve as the primary testbeds for emerging technologies. This is likely due to greater academic freedom, research funding availability and the pedagogical need to prepare students for a technology-driven workforce. In contrast, the slower adoption in primary and secondary education (3.4% and 13.5%, respectively) may be attributed to the more centralised and policy-driven nature of K-12 systems in many countries, particularly within European and Asian contexts, where curriculum changes and technology integration require broader systemic approval and teacher training initiatives.

Furthermore, the significant focus on STEM subjects (45.6%) aligns with global economic priorities and the international policy emphasis on developing a STEM-skilled workforce. However, this also raises questions from an international contextual perspective. The relative underrepresentation of humanities and social sciences could reflect a prevailing utilitarian view of AI in education, a trend that may be more dominant in the largely English-speaking research landscape captured in our review. Educational systems with different traditions, such as those in continental Europe that place a strong emphasis on the humanities and critical pedagogy, might offer alternative models for integrating AI to foster dialectical thinking rather than purely for skill acquisition.

These international variations suggest that while a technology like ChatGPT is global, its application is inherently local and contextual. The effectiveness of a universally available tool is ultimately mediated by regional educational philosophies, national policies and local classroom cultures. Therefore, educators and policymakers internationally should not seek a one-size-fits-all solution but rather adapt AI integration strategies to their specific cultural and pedagogical contexts, using findings like ours as a guide for reflection rather than a simple blueprint for implementation.

6 | Conclusions

This systematic review, guided by the GST framework, provides a comprehensive analysis of ChatGPT applications in education, addressing limitations in existing research. While

previous studies often focused on specific fields or education levels, our review of 92 articles published between November 2022 and March 2024 offers a holistic perspective across diverse educational settings, employing both the GST and KSA frameworks.

Our findings highlight ChatGPT's potential to enhance student learning experiences and outcomes, particularly in higher education (69.7%) and STEM disciplines (45.6%). The prevalence of timely feedback (55 mentions), content generation (20 mentions) and personalised learning (15 mentions) techniques across various GST elements underscores their effectiveness in improving learning outcomes and engagement. ChatGPT demonstrated positive impacts on knowledge construction (63 mentions) and skills development (59 mentions), with less emphasis on attitudinal outcomes (20 mentions). These findings suggest that ChatGPT is primarily employed to support cognitive and practical aspects of learning, aligning with its ability to provide personalised responses and engage in interactive discussions.

The analysis reveals that research on ChatGPT in education is heavily skewed towards higher education, with limited focus on primary (3.4%) and secondary education (13.5%). The dominance of web-based ChatGPT applications (80.4%) underlines the significance of online learning and the potential for AI technologies to be seamlessly integrated into educational platforms.

However, the review also identifies significant limitations in current research, including a predominance of small (40.4%) to medium-scale (43.8%) studies and limited investigation of attitudinal outcomes. These gaps highlight the need for future research to explore ChatGPT's potential across all educational levels, conduct large-scale investigations and examine its impact on learners' attitudes and behaviours. The limited representation of face-to-face (23.9%), experimental (16.9%) and informal learning environments (1.4%) indicates a need for broader research on ChatGPT's integration into various instructional settings.

The study's limitations, including the relatively small number of included studies due to ChatGPT's novelty, underscore the need for continued research in this rapidly evolving field. Additionally, while our analysis identified patterns in the distribution of ChatGPT applications across GST elements, we acknowledge that deeper exploration of the interrelationships between these elements (e.g., how subject-specific content influences specific learning outcomes) represents an important avenue for future research. Furthermore, this review focused exclusively on ChatGPT applications in education, without exploring other generative AI tools that might be particularly suited for specific disciplines such as STEM. As technology advances, more specialised AI tools with capabilities similar to or surpassing ChatGPT continue to emerge, potentially offering distinct advantages for different subject areas. Subsequent research should investigate the comparative effectiveness of various generative AI tools across different educational disciplines, identifying which tools are most appropriate for specific learning contexts and objectives. Research efforts should also focus on developing evidence-based frameworks and guidelines for effective ChatGPT interventions, considering the complex interplay between GST elements. This includes investigating

age-appropriate interventions for younger students, examining ChatGPT's potential to shape learners' dispositions and behaviours, and exploring its applications across a broader range of disciplines. More granular analysis of how specific subject content affects particular educational outcomes would provide valuable insights for practitioners seeking to implement ChatGPT in their specific educational contexts.

In conclusion, this review provides valuable insights into the current landscape of ChatGPT applications in education, extending beyond the scope of previous reviews. It sets the stage for future research to harness ChatGPT's full potential in transforming educational practices across diverse contexts, addressing the identified gaps in knowledge and contributing to a more comprehensive understanding of AI's role in education. By providing a nuanced understanding of how ChatGPT interacts with and influences various components of the educational ecosystem, this study informs evidence-based decision-making and guides future research in the rapidly evolving field of AI-enhanced education.

Author Contributions

Hsin-Yu Lee is the leader of this research, he is in charge of the research design, conducting teaching and learning experiment, data analysis and writing the manuscript. Yueh-Min Huang is responsible for designing research experiments, providing fundamental education theories and comments to this research, and he is also responsible for revising the manuscript. All authors spent more than 2 months to discuss and analyse the data. The author(s) read and approved the final manuscript. Ting-Ting Wu is responsible for assisting in the conduct of experiments and surveying related literature and proofreading the manuscript.

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Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** ejed70262-sup-0001-AppendixS1.docx.